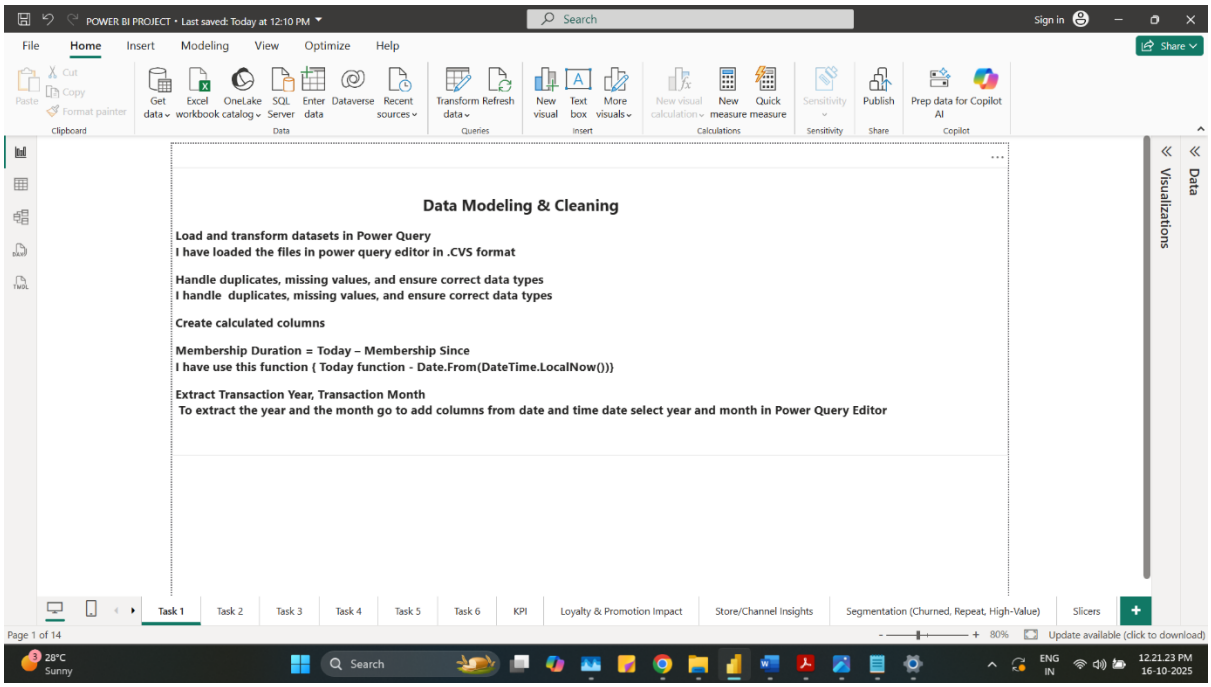
POWER BI PROJECT

Task 1



Data Modeling & Cleaning

Load and transform datasets in Power Query

I have loaded the files in power query editor in .CVS format

Handle duplicates, missing values, and ensure correct data types

I handle duplicates, missing values, and ensure correct data types

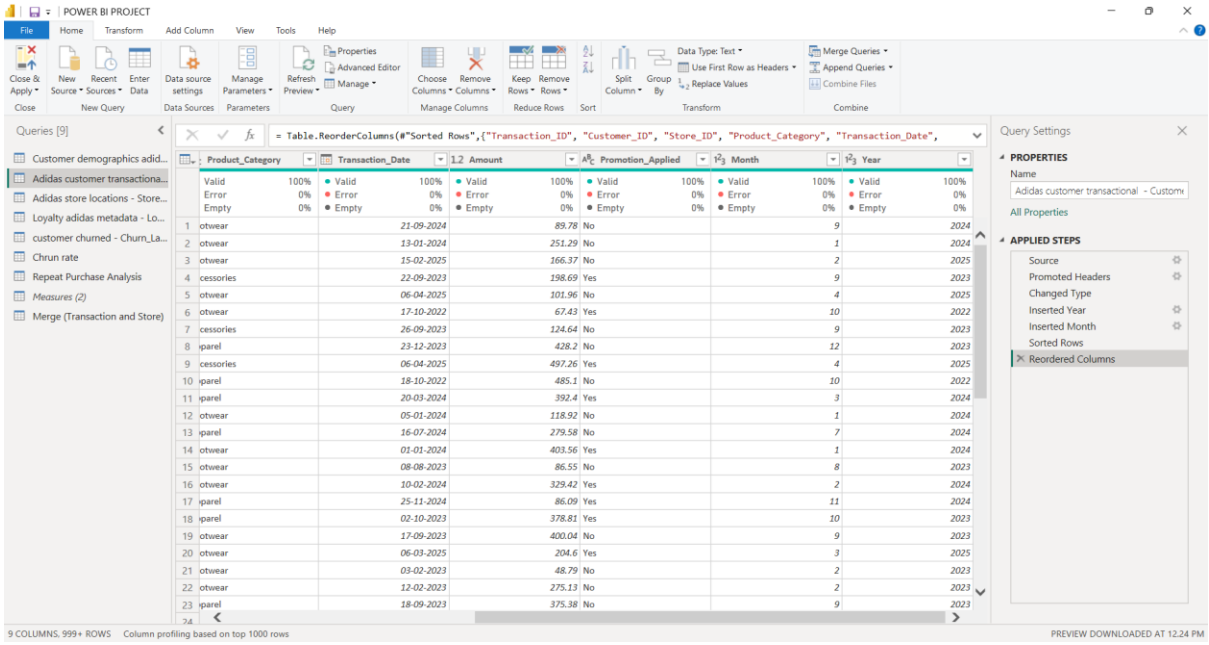
Create calculated columns

Membership Duration = Today – Membership Since

I have used this function {Today function - Date.From(DateTime.LocalNow())}

Extract Transaction Year, Transaction Month

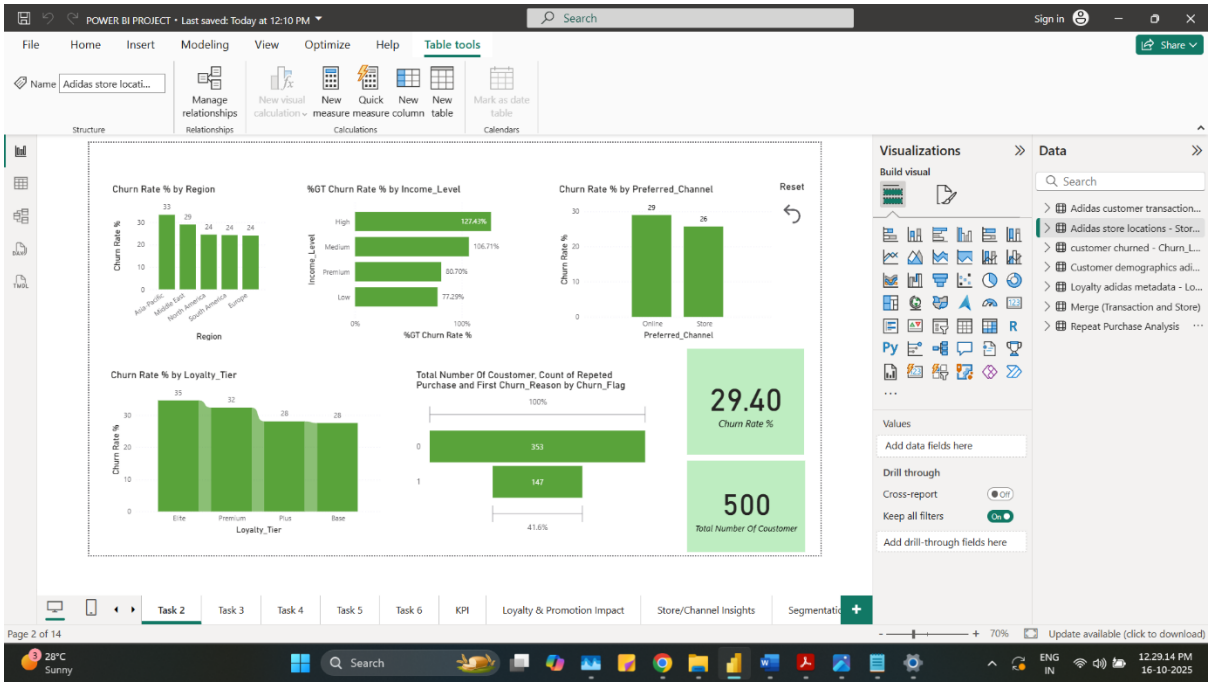
To extract the year and the month go to add columns from date and time dat e select year and month in Power Query Editor



Task 2

Create **Churn Rate KPI** = (Churned Customers / Total Customers) * 100

Churn Rate % = (CALCULATE(sum('customer churned - Churn_Labelled_Customers (1)'[Churn_Flag]))/[Total Number Of Customer]



Total Number Of Customer = COUNT('customer churned - Churn_Labelled_Customers (1)'[Customer_ID])

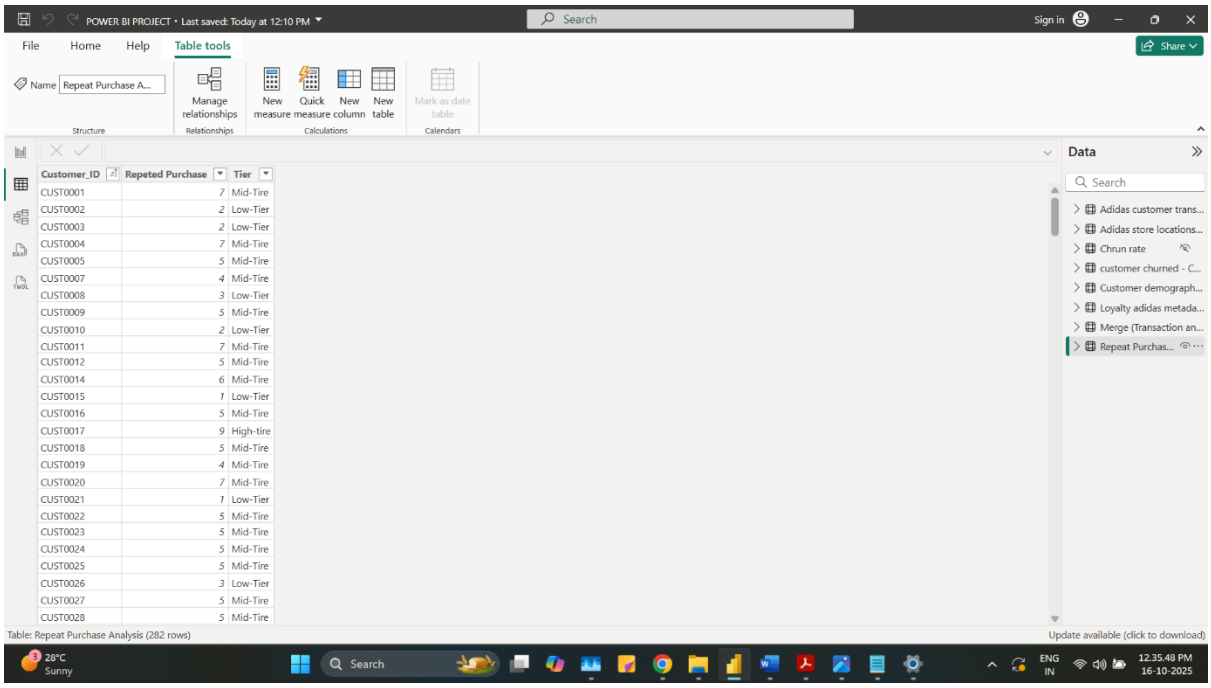
Key findings

The overall churn rate is 29.4%, meaning about one in three customers have stopped engaging. Interestingly, elite customers show the highest churn (35%), while base

members are the most loyal with the lowest churn (28%). Online shoppers (29%) are leaving slightly more than in-store ones (26%). High-income customers stand out with a very high churn rate (127%), suggesting possible dissatisfaction or that they're switching to competitors. Regionally, churn is highest in Asia-Pacific (33%) and lowest in Europe (24%).

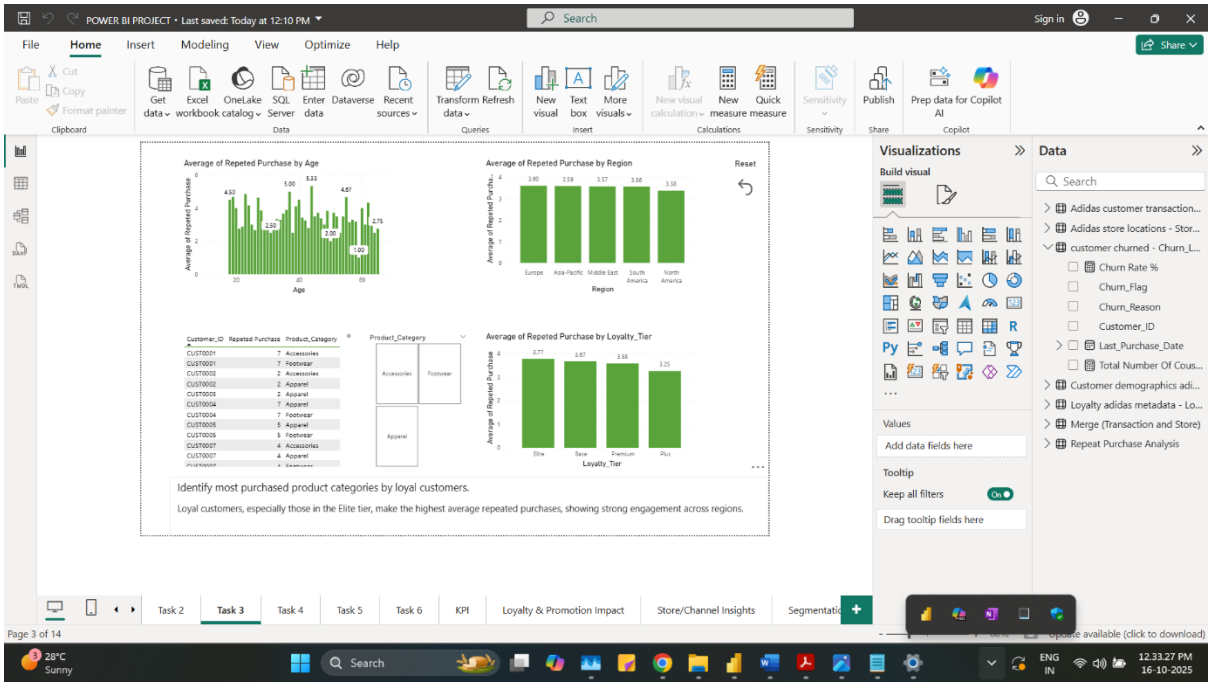
Overall, churn seems concentrated among elite, high-income, and online customers — meaning retention efforts should focus more on these key groups.

Task 3



Customer_ID	Repeted Purchase	Tier
CUST0001	7	Mid-Tire
CUST0002	2	Low-Tier
CUST0003	2	Low-Tier
CUST0004	7	Mid-Tire
CUST0005	5	Mid-Tire
CUST0007	4	Mid-Tire
CUST0008	3	Low-Tier
CUST0009	5	Mid-Tire
CUST0010	2	Low-Tier
CUST0011	7	Mid-Tire
CUST0012	5	Mid-Tire
CUST0014	6	Mid-Tire
CUST0015	1	Low-Tier
CUST0016	5	Mid-Tire
CUST0017	9	High-tire
CUST0018	5	Mid-Tire
CUST0019	4	Mid-Tire
CUST0020	7	Mid-Tire
CUST0021	1	Low-Tier
CUST0022	5	Mid-Tire
CUST0023	5	Mid-Tire
CUST0024	5	Mid-Tire
CUST0025	5	Mid-Tire
CUST0026	3	Low-Tier
CUST0027	5	Mid-Tire
CUST0028	5	Mid-Tire

I have done the repeated purchase analyses in power query editor.

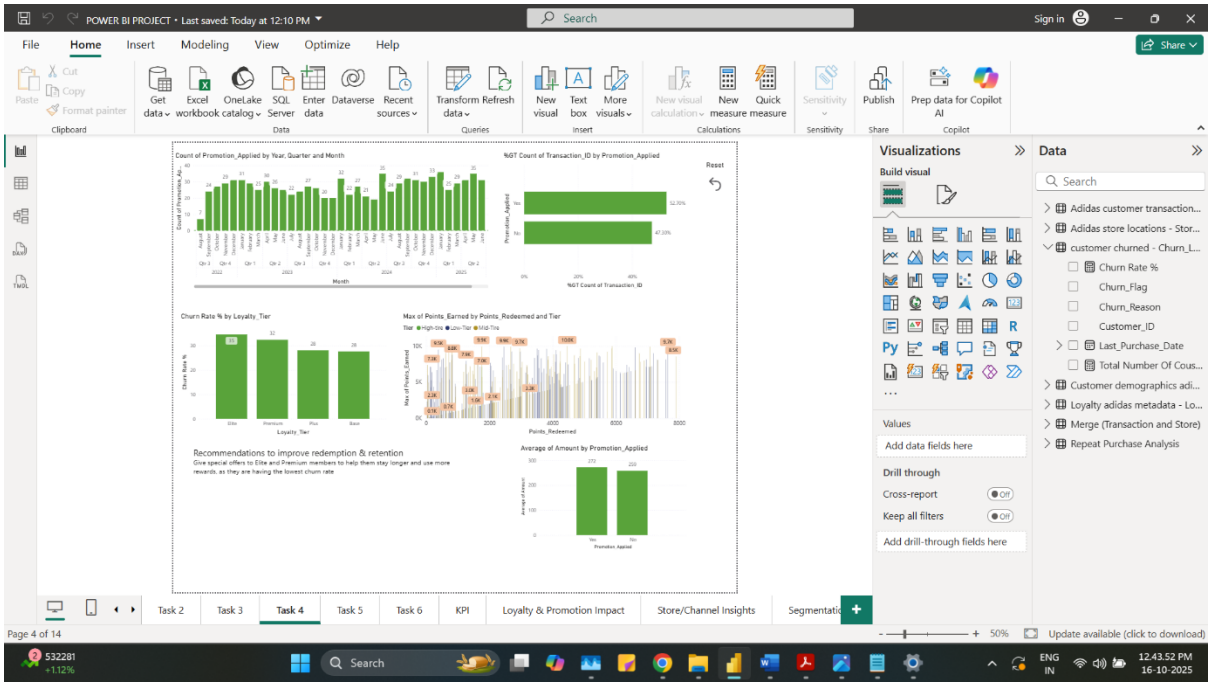


Key findings

Elite customers have the highest repeat purchases (3.77), showing strong brand loyalty. Accessories and Footwear are the top choices among loyal customers. Repeat purchases are slightly higher in Europe and Asia-Pacific (around 3.6), and younger customers aged 20–30 tend to buy more often.

Overall, younger and loyal customers — especially those in the Elite tier and in Europe or Asia-Pacific — show stronger repeat purchase behavior, mainly for Accessories and Footwear.

Task 4



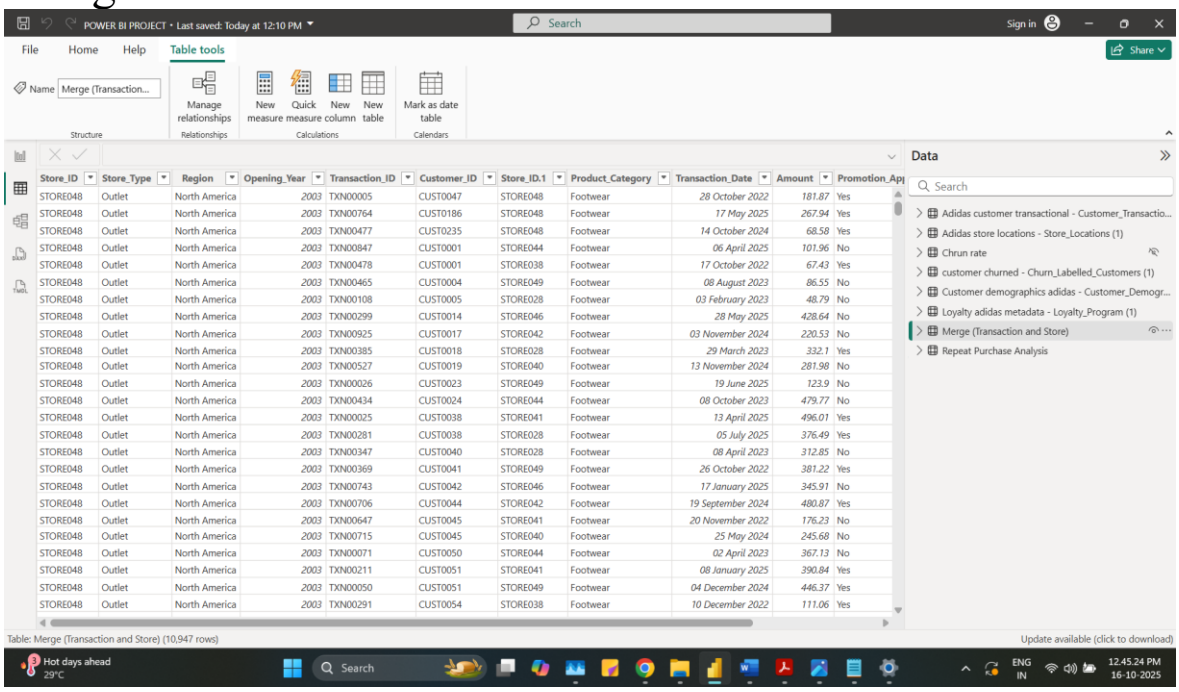
Key findings

About 52.7% of transactions were made using promotions, showing that offers play a big role in encouraging purchases. Elite and Premium members have the lowest churn rates (28–32%), making them the most loyal groups. Customers who redeem more points also tend to earn more rewards, especially in higher loyalty tiers. Promotions slightly lift average spending — ₹272 with offers compared to ₹259 without.

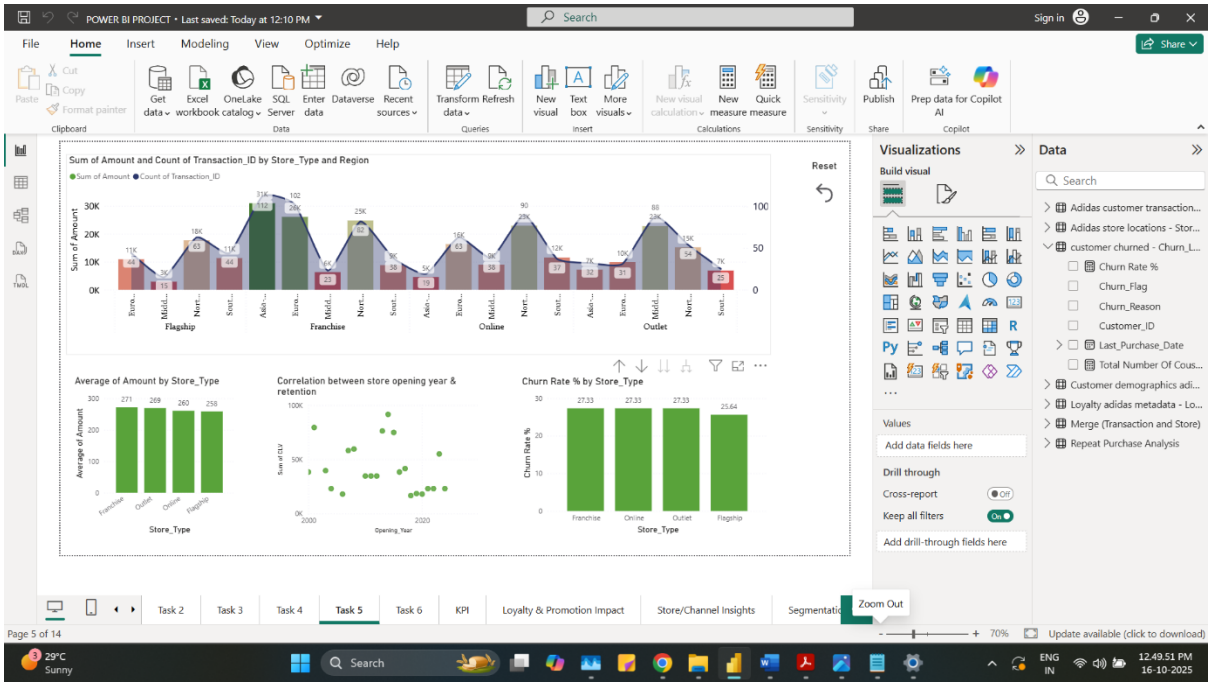
Overall, promotions and reward programs clearly help boost spending and retention, particularly among Elite and Premium members who respond well to deals and stay more loyal.

Task 5

Merge store data with transactions



Store_ID	Store_Type	Region	Opening_Year	Transaction_ID	Customer_ID	Store_ID.1	Product_Category	Transaction_Date	Amount	Promotion_Ap
STORE048	Outlet	North America	2003	TXN00005	CUST0047	STORE048	Footwear	28 October 2022	181.87	Yes
STORE048	Outlet	North America	2003	TXN00764	CUST0186	STORE048	Footwear	17 May 2025	267.94	Yes
STORE048	Outlet	North America	2003	TXN00477	CUST0235	STORE048	Footwear	14 October 2024	68.58	Yes
STORE048	Outlet	North America	2003	TXN00847	CUST0001	STORE044	Footwear	06 April 2025	101.96	No
STORE048	Outlet	North America	2003	TXN00478	CUST0001	STORE038	Footwear	17 October 2022	67.43	Yes
STORE048	Outlet	North America	2003	TXN00465	CUST0004	STORE049	Footwear	08 August 2023	86.55	No
STORE048	Outlet	North America	2003	TXN00108	CUST0005	STORE028	Footwear	03 February 2023	48.79	No
STORE048	Outlet	North America	2003	TXN00299	CUST0014	STORE046	Footwear	28 May 2025	428.64	Yes
STORE048	Outlet	North America	2003	TXN00925	CUST0017	STORE042	Footwear	03 November 2024	220.53	No
STORE048	Outlet	North America	2003	TXN00385	CUST0018	STORE028	Footwear	29 March 2023	332.1	Yes
STORE048	Outlet	North America	2003	TXN00527	CUST0019	STORE040	Footwear	13 November 2024	281.98	No
STORE048	Outlet	North America	2003	TXN00026	CUST0023	STORE049	Footwear	19 June 2025	123.9	No
STORE048	Outlet	North America	2003	TXN00434	CUST0024	STORE044	Footwear	08 October 2023	479.77	No
STORE048	Outlet	North America	2003	TXN00025	CUST0038	STORE041	Footwear	13 April 2025	496.01	Yes
STORE048	Outlet	North America	2003	TXN00281	CUST0038	STORE028	Footwear	05 July 2025	376.49	Yes
STORE048	Outlet	North America	2003	TXN00347	CUST0040	STORE028	Footwear	08 April 2023	312.85	No
STORE048	Outlet	North America	2003	TXN00369	CUST0041	STORE049	Footwear	26 October 2022	381.22	Yes
STORE048	Outlet	North America	2003	TXN00743	CUST0042	STORE046	Footwear	17 January 2025	345.91	No
STORE048	Outlet	North America	2003	TXN00706	CUST0044	STORE042	Footwear	19 September 2024	480.87	Yes
STORE048	Outlet	North America	2003	TXN00647	CUST0045	STORE041	Footwear	20 November 2022	176.23	No
STORE048	Outlet	North America	2003	TXN00715	CUST0045	STORE040	Footwear	25 May 2024	245.68	No
STORE048	Outlet	North America	2003	TXN00071	CUST0050	STORE044	Footwear	02 April 2023	367.13	No
STORE048	Outlet	North America	2003	TXN00211	CUST0051	STORE041	Footwear	08 January 2025	390.84	Yes
STORE048	Outlet	North America	2003	TXN00050	CUST0051	STORE049	Footwear	04 December 2024	446.37	Yes
STORE048	Outlet	North America	2003	TXN00291	CUST0054	STORE038	Footwear	10 December 2022	111.06	Yes



Key findings

Flagship stores and outlets record the highest number of transactions and overall sales, especially in Europe and South America. Online stores have fewer transactions but still manage to bring in steady revenue.
Average spending per transaction is fairly consistent across store types, ranging from ₹258 to ₹271, with franchise stores slightly ahead. The scatter plot comparing store opening year and retention shows no clear pattern — meaning the store's age doesn't really affect retention.
Churn rates are also quite similar across all store types, around 27%, though flagship stores perform a bit better with a slightly lower churn of 25.6%.

Overall, the store's age doesn't seem to impact retention, but flagship stores stand out for maintaining better customer loyalty and higher transaction values.

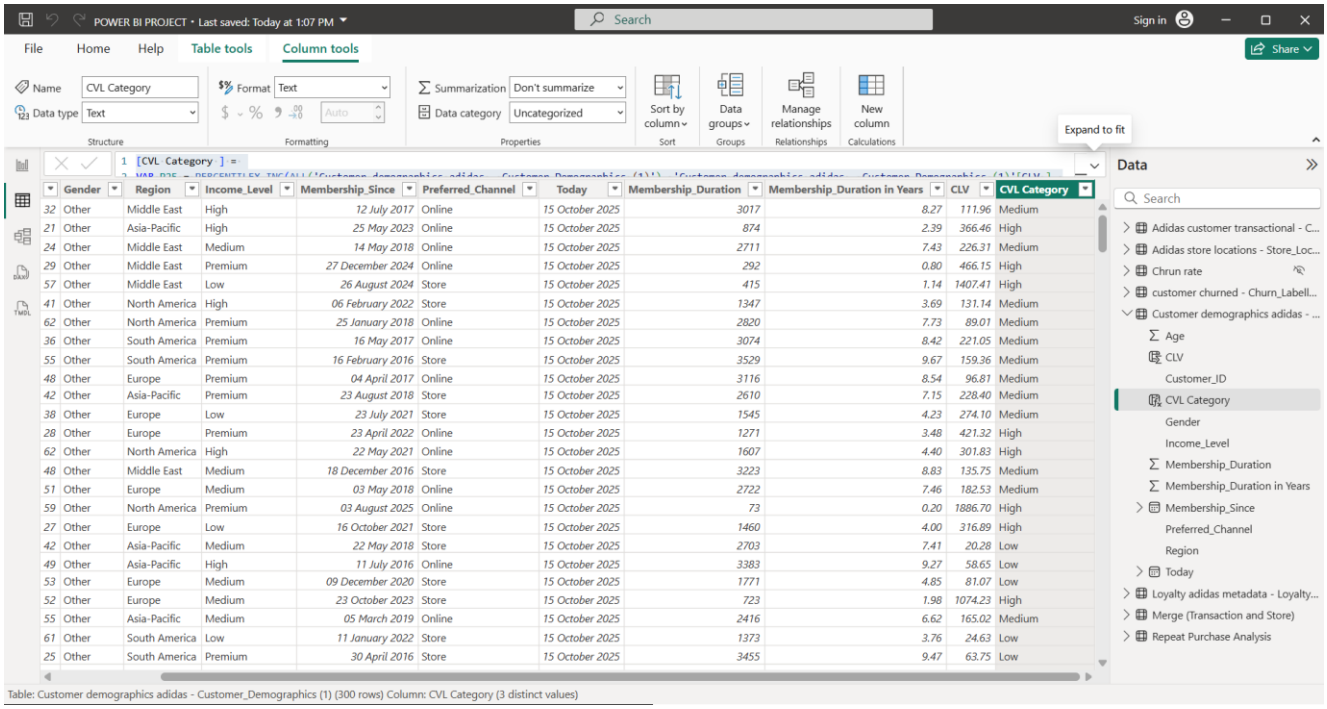
Task 6

CLV = Total Amount Spent / Membership Duration (Years)

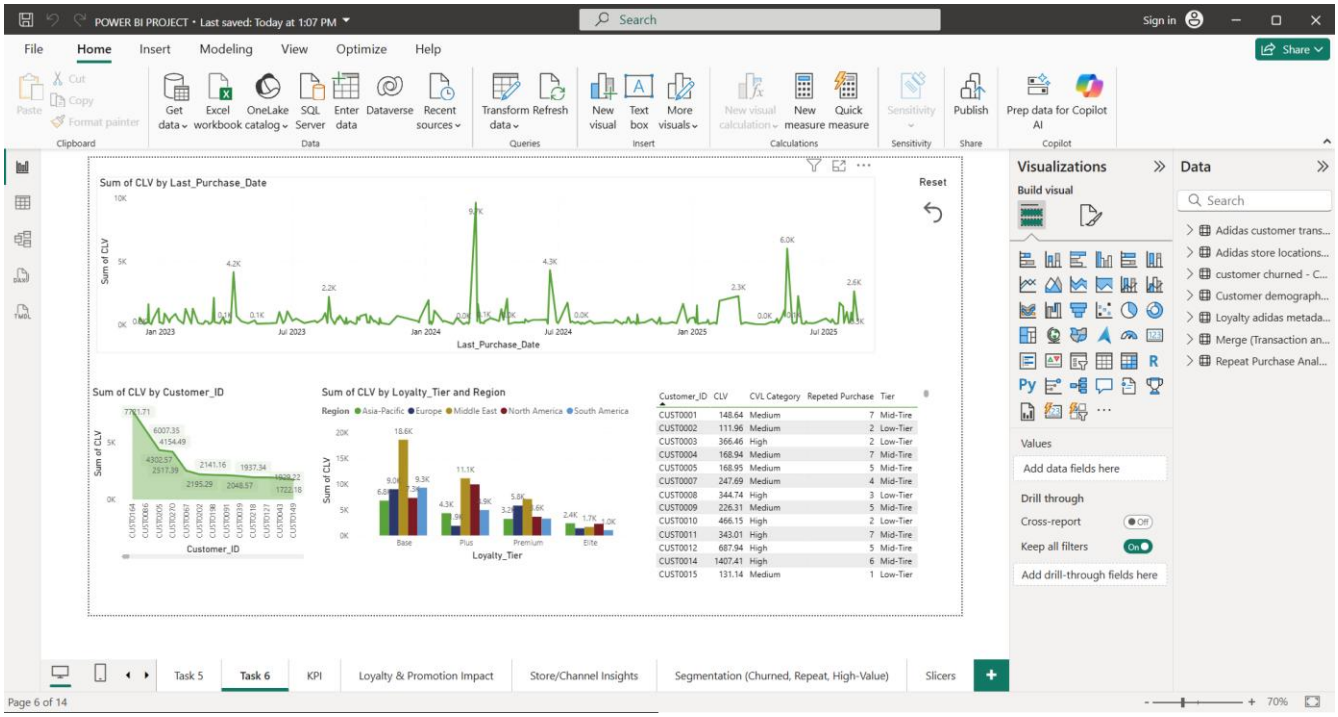
[CLV] = DIVIDE (CALCULATE (SUM ('Adidas customer transactional - Customer_Transactions (1)'[Amount])),'Customer demographics adidas - Customer_Demographics (1)'[Membership_Duration in Years])

Segment customers into **Low, Medium, High CLV**

```
[CVL Category ] =
VAR P25 = PERCENTILEX.INC(ALL('Customer demographics adidas - Customer_Demographics (1)'), 'Customer demographics adidas - Customer_Demographics (1)'[CLV ], 0.25)
VAR P70 = PERCENTILEX.INC(ALL('Customer demographics adidas - Customer_Demographics (1)'), 'Customer demographics adidas - Customer_Demographics (1)'[CLV ], 0.70)
VAR CVL = 'Customer demographics adidas - Customer_Demographics (1)'[CLV ]
RETURN
    SWITCH(
        TRUE(),
        CVL < P25, "Low",
        CVL < P70, "Medium",
        "High"
    )
```



	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Today	Membership_Duration	Membership_Duration in Years	CLV	CVL Category
32	Other	Middle East	High	12 July 2017	Online	15 October 2025	3017	8.27	111.96	Medium
21	Other	Asia-Pacific	High	25 May 2023	Online	15 October 2025	874	2.39	366.46	High
24	Other	Middle East	Medium	14 May 2018	Online	15 October 2025	2711	7.43	226.31	Medium
29	Other	Middle East	Premium	27 December 2024	Online	15 October 2025	292	0.80	466.15	High
57	Other	Middle East	Low	26 August 2024	Store	15 October 2025	415	1.14	1407.41	High
41	Other	North America	High	06 February 2022	Store	15 October 2025	1347	3.69	131.14	Medium
62	Other	North America	Premium	25 January 2018	Online	15 October 2025	2820	7.73	89.01	Medium
36	Other	South America	Premium	16 May 2017	Online	15 October 2025	3074	8.42	221.05	Medium
55	Other	South America	Premium	16 February 2016	Store	15 October 2025	3529	9.67	159.36	Medium
48	Other	Europe	Premium	04 April 2017	Online	15 October 2025	3116	8.54	96.81	Medium
42	Other	Asia-Pacific	Premium	23 August 2018	Store	15 October 2025	2610	7.15	228.40	Medium
38	Other	Europe	Low	23 July 2021	Store	15 October 2025	1545	4.23	274.10	Medium
28	Other	Europe	Premium	23 April 2022	Online	15 October 2025	1271	3.48	421.32	High
62	Other	North America	High	22 May 2021	Online	15 October 2025	1607	4.40	301.83	High
48	Other	Middle East	Medium	18 December 2016	Store	15 October 2025	3223	8.83	135.75	Medium
51	Other	Europe	Medium	03 May 2018	Online	15 October 2025	2722	7.46	182.53	Medium
59	Other	North America	Premium	03 August 2025	Online	15 October 2025	73	0.20	1886.70	High
27	Other	Europe	Low	16 October 2021	Store	15 October 2025	1460	4.00	316.89	High
42	Other	Asia-Pacific	Medium	22 May 2018	Store	15 October 2025	2703	7.41	20.28	Low
49	Other	Asia-Pacific	High	11 July 2016	Online	15 October 2025	3383	9.27	58.65	Low
53	Other	Europe	Medium	09 December 2020	Store	15 October 2025	1771	4.85	81.07	Low
52	Other	Europe	Medium	23 October 2023	Store	15 October 2025	723	1.98	1074.23	High
55	Other	Asia-Pacific	Medium	05 March 2019	Online	15 October 2025	2416	6.62	165.02	Medium
61	Other	South America	Low	11 January 2022	Store	15 October 2025	1373	3.76	24.63	Low
25	Other	South America	Premium	30 April 2016	Store	15 October 2025	3455	9.47	63.75	Low



Customer_ID	CLV	CVL Category	Repeted Purchase	Tier
CUST0001	148.64	Medium	7	Mid-Tire
CUST0002	111.96	Medium	2	Low-Tier
CUST0003	366.46	High	2	Low-Tier
CUST0004	168.94	Medium	7	Mid-Tire
CUST0005	168.95	Medium	5	Mid-Tire
CUST0007	247.69	Medium	4	Mid-Tire
CUST0008	344.74	High	3	Low-Tier
CUST0009	226.31	Medium	5	Mid-Tire
CUST0010	466.15	High	2	Low-Tier
CUST0011	343.01	High	7	Mid-Tire
CUST0012	687.94	High	5	Mid-Tire
CUST0014	1407.41	High	6	Mid-Tire
CUST0015	131.14	Medium	1	Low-Tier

Key findings

The line chart shows clear ups and downs in total CLV between January 2023 and January 2025, with noticeable peaks during certain months — likely due to seasonal shopping trends or successful promotions that boosted spending.

Looking at loyalty tiers, Premium members have the highest CLV (around ₹18.6K), followed by Plus (₹11.1K). Base-tier customers have much lower CLV, suggesting room to strengthen loyalty engagement in this group.

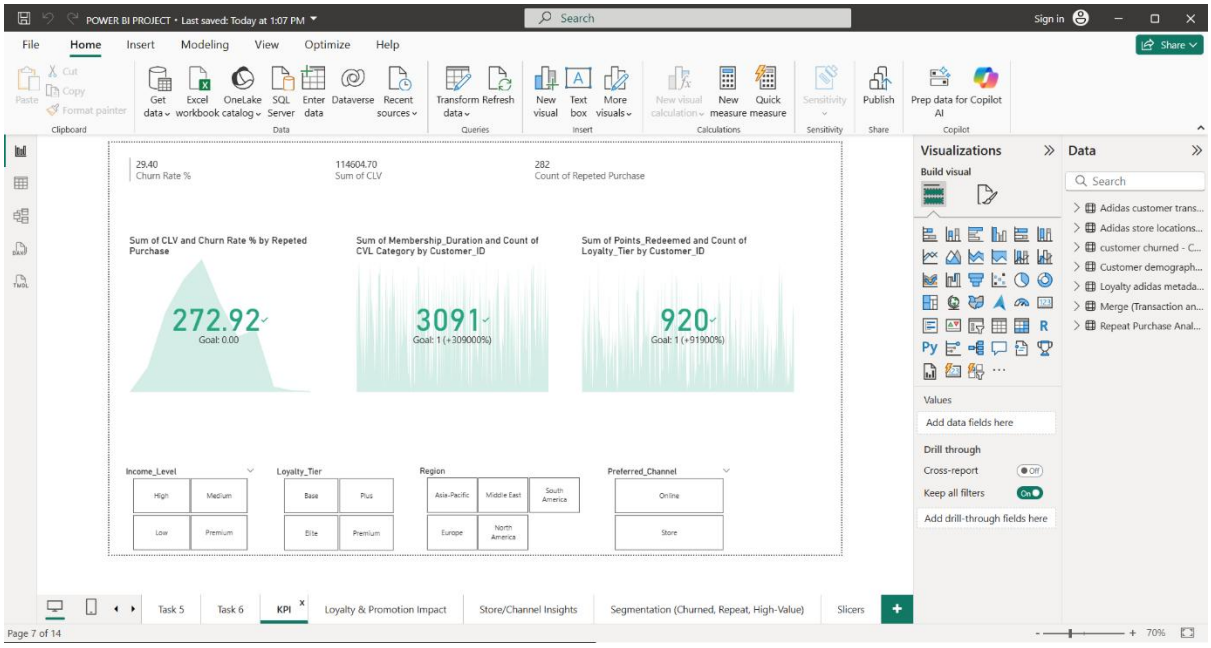
Regionally, CLV varies across locations, with Asia-Pacific showing higher averages, creating good opportunities for targeted marketing in that region.

The customer details table highlights loyalty categories and purchase frequency, helping identify high-value customers worth focusing retention efforts on — especially those in higher tiers who make frequent repeat purchases.

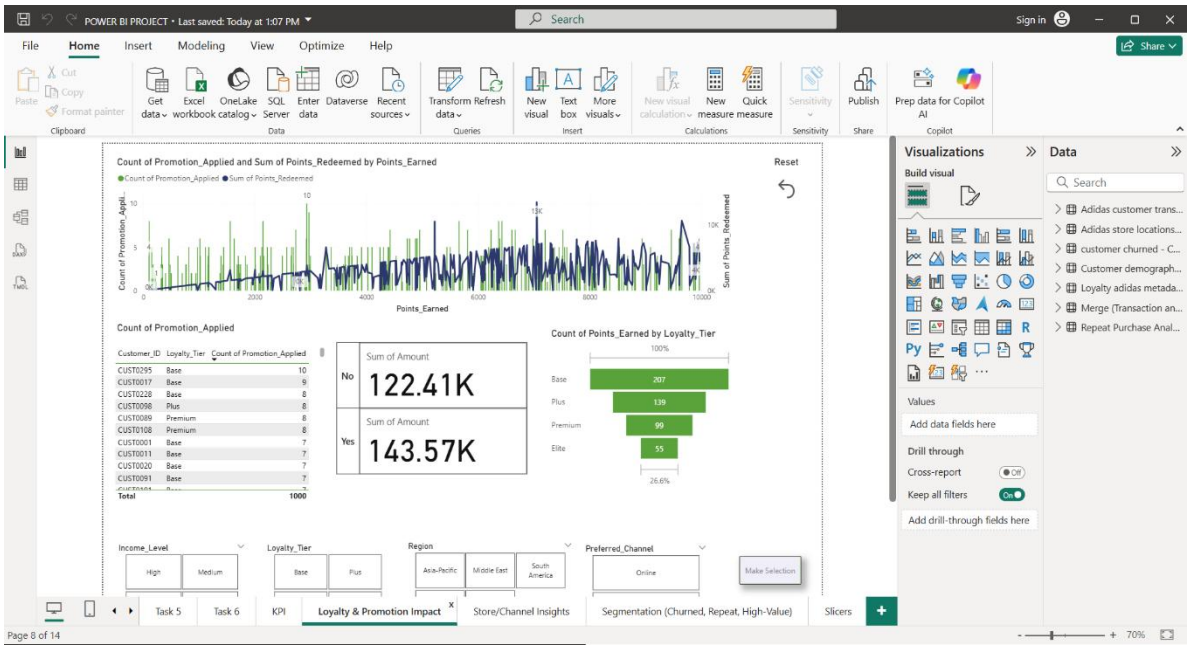
Overall, these insights reveal clear opportunities to grow revenue through targeted marketing, improved loyalty strategies, and deeper engagement with high-value customers across tiers and regions.

Task 7

1. KPI



2. Loyalty & Promotion Impact

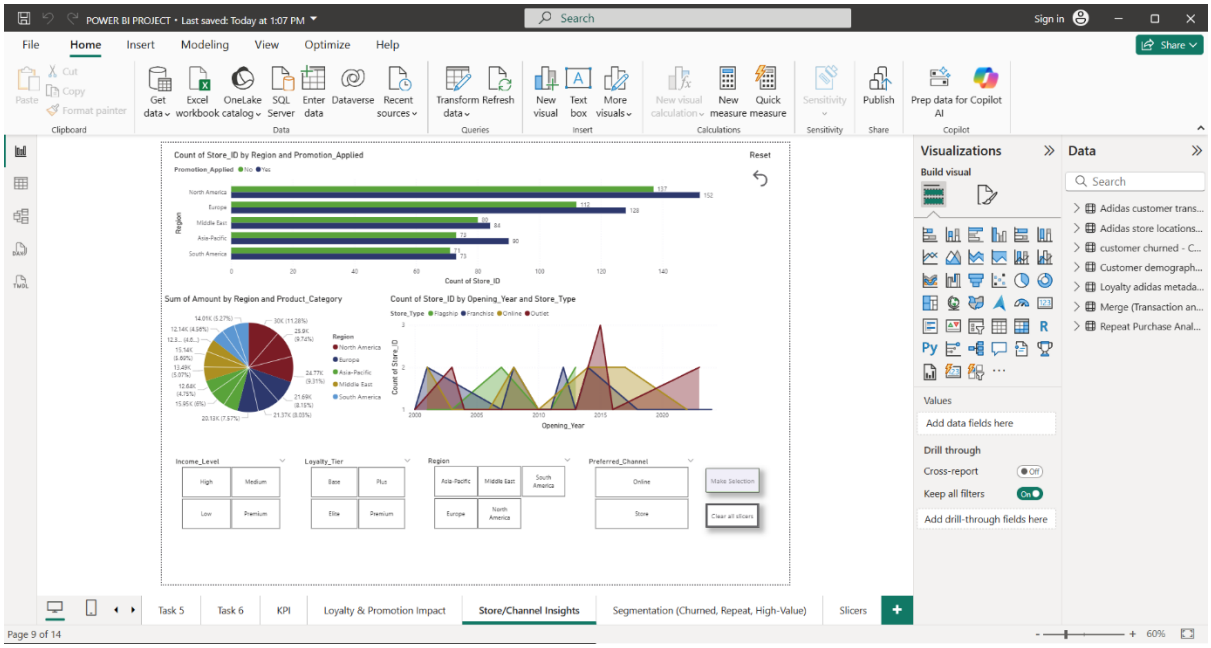


Key findings

Customers who used promotions spent noticeably more — ₹143.57K compared to ₹122.41K for those who didn't — showing that promotions clearly boost spending.

When looking at loyalty tiers, Base members earned the most points (207), followed by Plus (139), Premium (99), and Elite (55). This means most engagement is coming from lower-tier customers.
The line chart shows steady activity between points earned, promotions applied, and points redeemed, indicating consistent participation in loyalty programs. Interestingly, Base and Premium members are the most frequent users of promotions.
Overall, promotional campaigns are driving both higher spending and engagement, especially among Base-tier customers. However, Elite members seem less involved in loyalty activities, which could be an area to improve.

3. Store/Channel Insights



Key findings

The bar charts show that store openings vary by region, with North America leading in total store count.

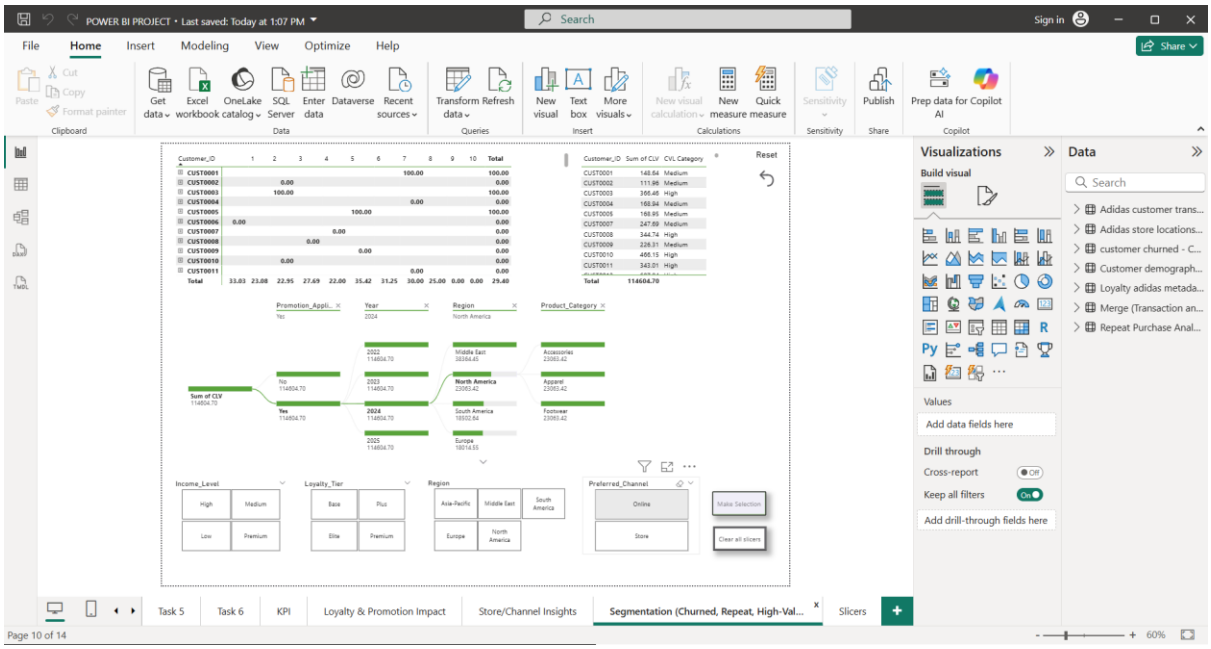
Promotions clearly impact store performance — stores running promotions tend to perform noticeably better than those that don't.

The pie chart breaks down sales by region and product category, revealing which areas and categories bring in the most revenue.

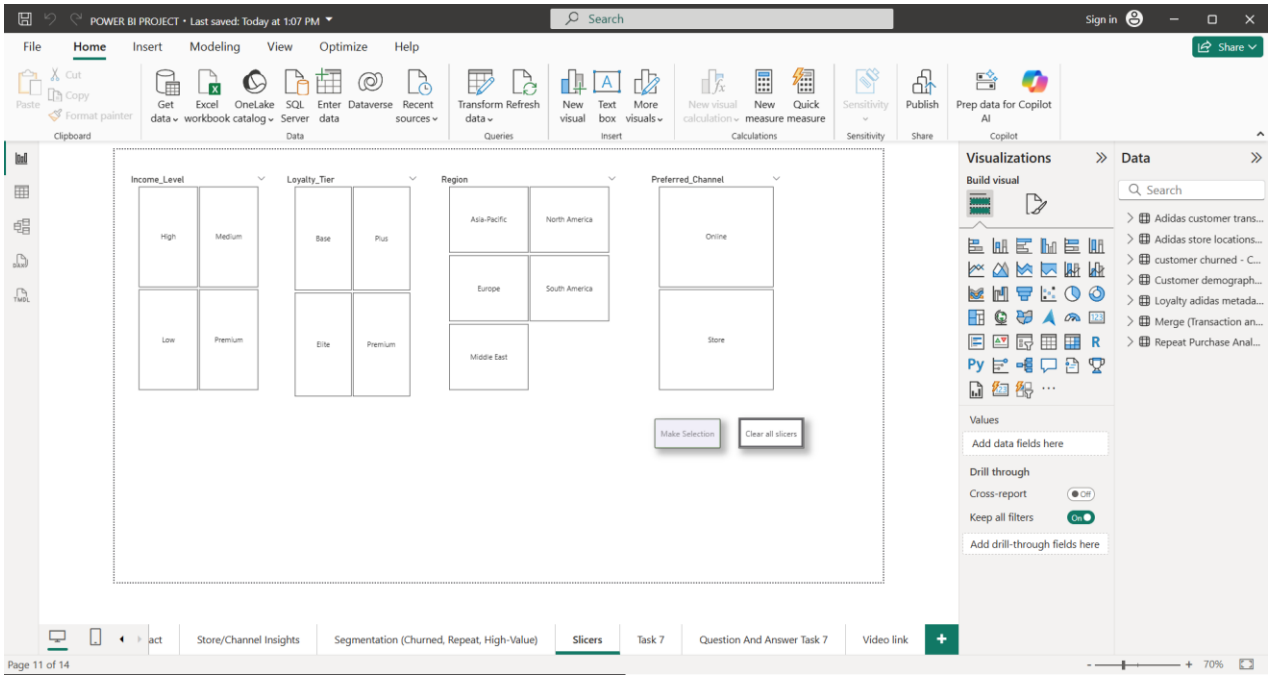
The line chart highlights store opening trends over the years, with sharp increases in certain years, likely reflecting strategic expansion or new market opportunities.

Customer filters such as income level and loyalty tier help identify key segments for targeted marketing.

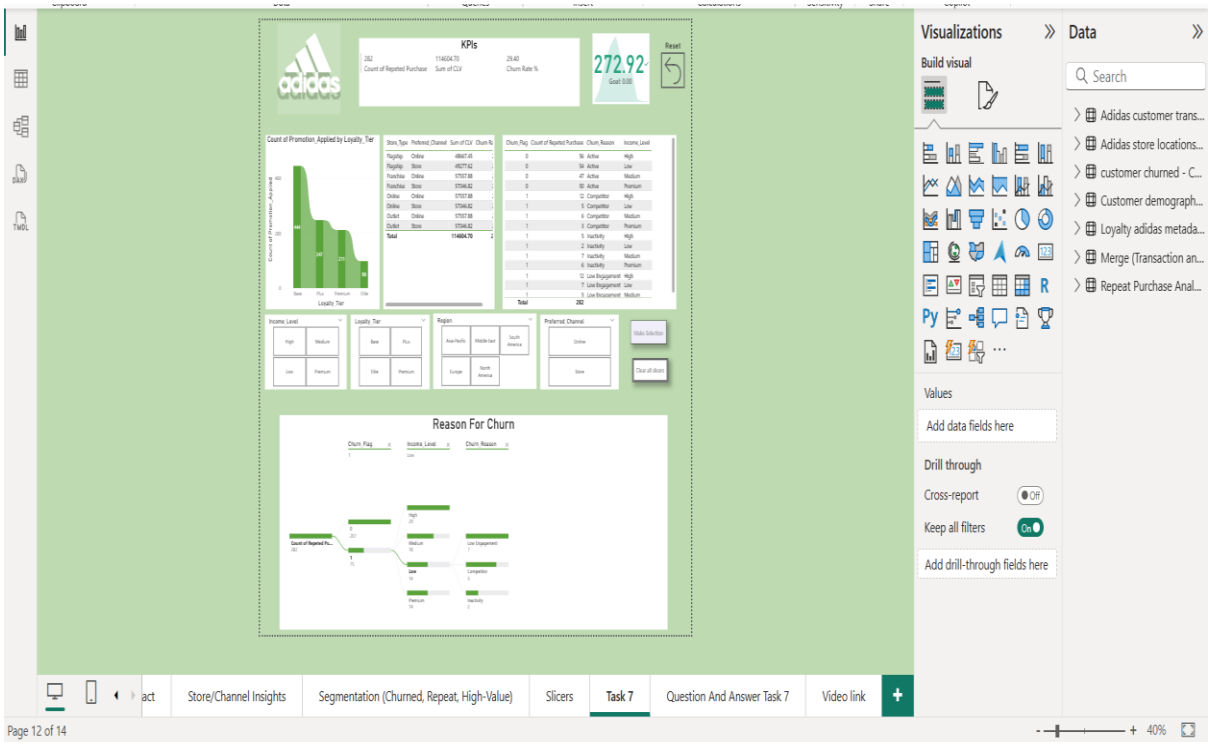
4. Segmentation



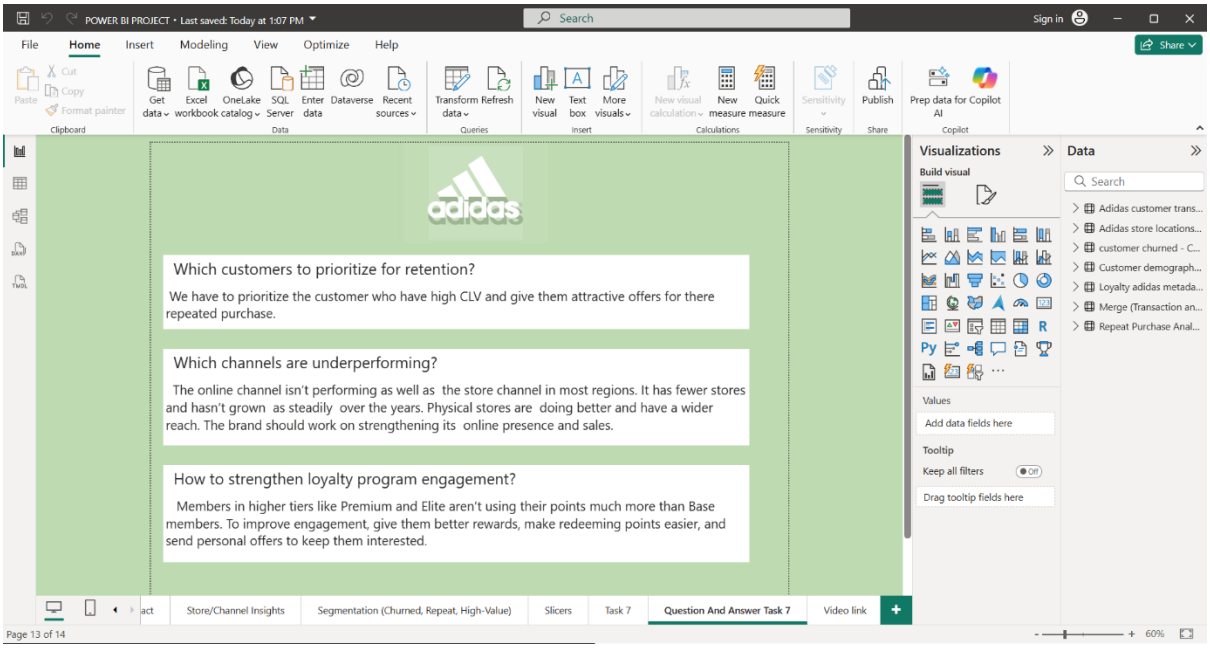
5. Slicers: Region, Channel, Income, Loyalty Tier



Final Dashboard



6. Top 3 recommendations for Adidas: (Question and Answer)



Which customers to prioritize for retention?

We have to prioritize the customer who have high CLV and gi ve them attractive offers for there repeated purchase.

Which channels are underperforming?

The online channel isn't performing as well as the store channel in most regions. It has fewer stores and has n't grown as steadily over the years. Physical stores are doin g better and have a wider reach. The brand should work on s trengthening its online presence and sales.

How to strengthen loyalty program engagement?

Members in higher tiers like Premium and Elite aren't using t heir points much more than Base members. To improve enga gement, give them better rewards, make redeeming points ea sier, and send personal offers to keep them interestd.

Video Link -

https://www.loom.com/share/6993b7de980b4169b5463b9a6afb630d?sid=546d1284-d2d3-4936-85ef-7b19ed40fc31